The NEAR FUTURE Of ELECTRONICS

So far, electronics has depended heavily on silicon for its development. But now new materials and techniques are emerging to take electronics further

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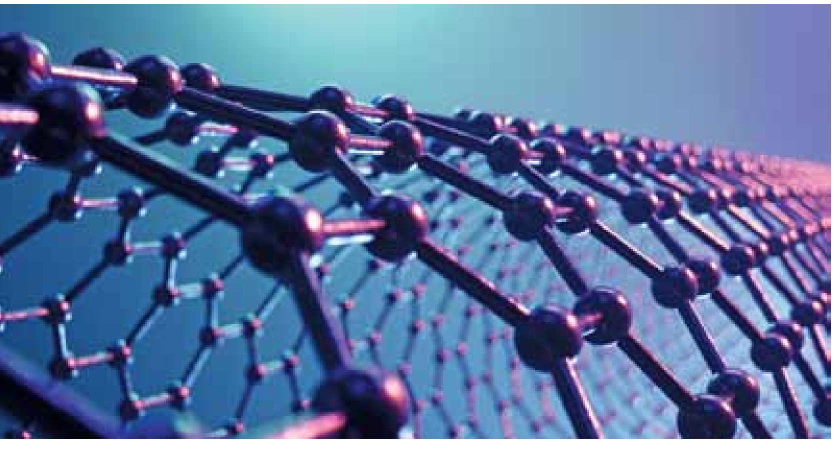
The era of electronics started much before it got its name as electronics. Earlier it was called radio engineering. The electronics word came into vogue in late 1940 and became prominent by late 1950's. The electronics industry was revolutionised by the invention of first transistor in 1948, which was a germanium based semiconductor. However, germanium has the drawback of breaking down at around 80°C, so most of it is now silicon.

Now, for more than seven decades, we have been living with electronics hand in hand. Our reliance and dependency on it are increasing day by day and we have come to an era where without electronics we cannot survive.

For decades and decades together, silicon has remained the only option for electronics. And we have seen the performance of silicon based devices in terms of speed, power consumption, size, and cost improving multi-fold. But recent developments in material science is paving new ways to electronics through fields such as:

- 2D electronics
- Organic electronics
- Memristors
- Spintronics
- Molecular electronics



2D electronics

In the year 2010, Nobel Prize for Physics was awarded to Andre Geim and Konstantin Novoselov for the experiments carried out on graphene, which is a structural variant of carbon. Carbon atoms in graphene form a hexagonal two-dimensional structure (lattice), which is as thick as atomic layer and has a high electrical conductivity, high thermal conductivity, high mechanical flexibility, and probably the highest mechanical strength (strongest material tested ever). Graphene is from where the concept of two-dimensional (2D) electronics started.

A 2D semiconductor used in two-dimensional electronics has its thickness in atomic scale. It has only length and breadth, as its third dimension of thickness is almost missing. This semiconductor can conduct more electricity and heat, consumes lesser power, and exhibits greater mechanical strength and stability.

2D materials are incredibly thin and lightweight and are stable at size below 10nm ($1nm = 10^{-9}$ metre), the point at which traditional semiconductor material start failing. Graphene is also highly transparent to the incident light, and it absorbs only 2% of the light incident upon it.

Graphene poses some technical challenges while being used in digital electronics (because at atomic level it has lack of band gap). Going forward into the domain of material science, nano layers of material of the same group—such as Si, Ge,